

MarkItDown is a lightweight Python utility for converting various files to Markdown for use with LLMs and related text analysis pipelines. To this end, it is most comparable to [textextract](#), but with a focus on preserving important document structure and content as Markdown (including: headings, lists, tables, links, etc.) While the output is often reasonably presentable and human-friendly, it is meant to be consumed by text analysis tools -- and may not be the best option for high-fidelity document conversions for human consumption.

MarkItDown currently supports the conversion from:

- PDF
- PowerPoint
- Word
- Excel
- Images (EXIF metadata and OCR)
- Audio (EXIF metadata and speech transcription)
- HTML
- Text-based formats (CSV, JSON, XML)
- ZIP files (iterates over contents)
- YouTube URLs
- EPubs
- ... and more!

Why Markdown?

Markdown is extremely close to plain text, with minimal markup or formatting, but still provides a way to represent important document structure. Mainstream LLMs, such as OpenAI's GPT-4o, natively "*speak*" Markdown, and often incorporate Markdown into their responses unprompted. This suggests that they have been trained on vast amounts of Markdown-formatted text, and understand it well. As a side benefit, Markdown conventions are also highly token-efficient.

Prerequisites

MarkItDown requires Python 3.10 or higher. It is recommended to use a virtual environment to avoid dependency conflicts.